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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/576,907	01/05/2024	Keith Sullivan	AFLIG-172-PCT-US	1958

22827 7590 03/25/2026

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EXAMINER
KIANNI, KAVEH C

ART UNIT	PAPER NUMBER
2874	

NOTIFICATION DATE	DELIVERY MODE
03/25/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/576,907	Applicant(s) Sullivan, Keith	
	Examiner KAVEH C KIANNI	Art Unit 2874	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 2/10/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) 9-18 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 1/5/24 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Detailed Action

- Applicant's election with traverse of Group I and species figs. 1 and 11 in response/amendment is acknowledged. The traverse is on the grounds that the embodiments related/define the same invention as depicted in fig. 1 and that the search can be done without serious burden. This is not found persuasive as each invention would require different search subclasses and different search strings with significant burden. The non-elected claims 9-17, now dependent on claim 1, are tentatively withdrawn, but maybe adjoin by the examiner at the time of allowance. The requirement is still deemed proper and is therefore made FINAL.

Claim Rejections - 35 USC 112

The following is a quotation of 35 U.S.C. 112(b):

(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:

The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

Claims 1 and its dependent claims are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as failing to set forth the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant(s) regard as their invention.

Claim 1 is indefinite for reciting first pin to pin distance of first interfacing of approximately 4.6 millimeters and a first total terminal spacing of approximately 2.75 millimeters, and second interface, the second interface with a second pin to pin distance of approximately 5.3 millimeters and a second total terminal spacing of approximately 3.75 millimeters; as approximately is an indefinite limitation for the scope of the cited claims and may be correct using a more precise language such as 'substantially'.
All dependent claims depending to claim 1 are rejected because of dependency.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries set forth in Graham v. John Deere Co., 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

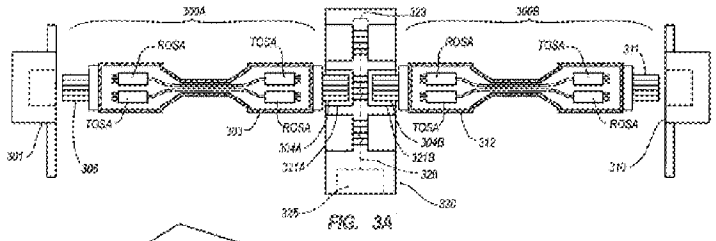
In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

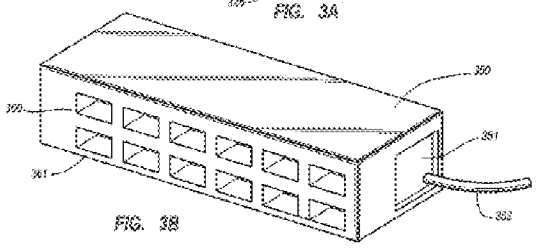
Claims 1-8 are rejected under AIA 35 U.S.C. 103(a) as being unpatentable over "Aronson" et al. US 20070237470 A1.

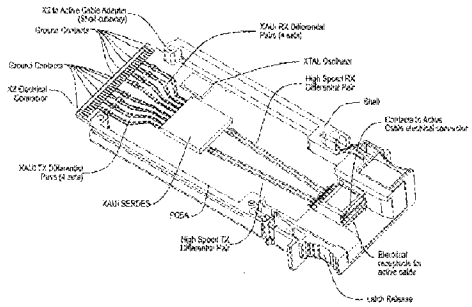
With regard to claim 1, Aronson teaches multi-fiber connector apparatus, the apparatus (see figs. 1-24; abstract, summary) comprising:

a housing surrounding a first multi-fiber connector and a second multi-fiber connector (shown in at least figs. 3, 13 parag. 01-04; also related 22, 1-3 and 16-18…), the housing comprising at least two first multi-fiber connectors operably coupled to the

second multi-fiber connector (shown in at least figs. 3, 13, parag. 0114, …)







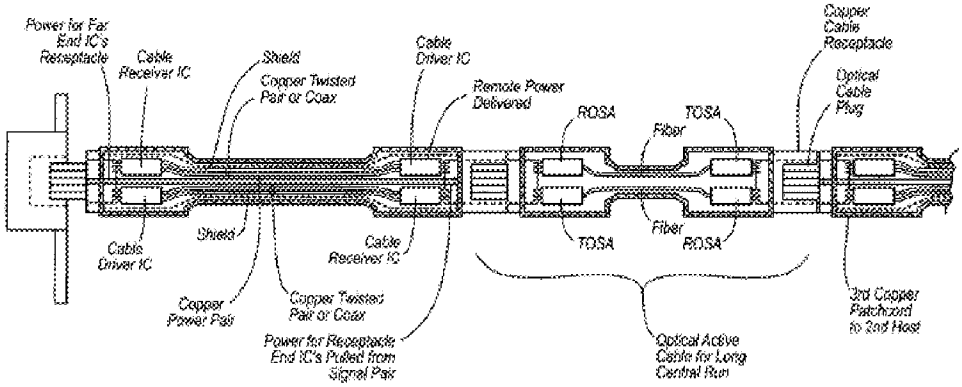


FIG. 13A

the first multi-fiber connectors operably coupled to respective second multi-fiber connectors by a plurality of optical fibers (i.e., 131, 132) within the housing (shown in at least figs. 3, 13, parag. 0114, …), wherein the first multi-fiber connector comprises a first interface, the first interface comprising a first pin to pin distanced from each other, and wherein the second multi-fiber connector comprises a second interface, the second interface comprising a second pin to pin distance of approximately terminal spacing (clearly shown in at least figs. 3 and 13 with not labeled pin interfacing).

However, Aronson does is silent on the above first pin to pin distance of first interfacing of approximately 4.6 millimeters and a first total terminal spacing of approximately 2.75 millimeters, and second interface, the second interface with a second pin to pin distance of approximately 5.3 millimeters and a second total terminal spacing of approximately 3.75 millimeters. Nonetheless, Aronson states that the width of the connector width is less than ½ inch (or >12 mm) and as shown in the figures the pin distances/spacings in the order of a few millimeters. Thus, It would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention was made to modify the pins spacings from each other in a span of millimeters from each other to achieve a high speed data transmission (see parag. 0117) as it has been held that where the general conditions of a claim are disclosed in the prior art, discovering the optimum or working ranges involves only routine skill in the art. In re Aller, 105 USPQ 233.

- The statements advanced in rejection of claim 1, above, as to the applicability and disclosure of and the motivation are incorporated herein in rejection of the following claims as follows:

With regard to claim 2-3 though, Aronson does not explicitly state wherein the first multi-fiber connector comprises a twelve-way multi-fiber connector and eight optical fibers operably coupled to the twelve-way multi-fiber connector, wherein the first multi-fiber connector comprises a twelve-way factor multi-fiber connector and factor-of-eight optical fibers operably coupled to the twelve-way factor multi-fiber connector.

Nonetheless, Aronson states that the number of fibers is determined by the by the optical link budget to which the E-O cable and optical transceiver comply and that by the optical link budget to which the E-O cable and optical transceiver comply (see motivation stated in claim 1.

With regard to claim 4, 'wherein the first multi-fiber connector comprises a first key interface, and wherein the second multi-fiber connector comprises a second key interface different from the first key interface' nonetheless, Aronson states that different conventional connectors can be used in the invention (parag. 0076) and that such keys in connectors are extremely conventional, i.e., US 20230266546 A1, so as to perform precise optical alignment.

With regard to claims 5-8 The multi-fiber connector apparatus of claim 4, wherein the second multi-fiber connector comprises a sixteen-way factor multi-fiber connector (see similar arguments for claim 2-3 is applicable in rejection of this claim).

6. (Original) The multi-fiber connector apparatus of claim 4, wherein the sixteen-way factor multi-fiber connector comprises any one or more pluralities of optical fibers (see similar arguments for claim 2-3 is applicable in rejection of this claim).

7. (Original) The multi-fiber connector apparatus of claim 4, wherein the second multi-fiber connector comprises a sixteen-way multi-fiber connector (see similar arguments for claim 2-3 is applicable in rejection of this claim).

8. (Original) The multi-fiber connector apparatus of claim 4, wherein the eight optical fibers at each first multi-fiber connector are operably coupled within the housing to respective holes at the second-multi-fiber connector (see figs. 2-6 and 13-17 and see similar arguments for claim 2-3 is applicable in rejection of this claim).

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Citation of Relevant Prior Art

Prior art made of record and not relied upon is considered pertinent to applicant's disclosure. In accordance with MPEP 707.05 the following references are pertinent in rejection of this application since they provide substantially the same information disclosure as this patent does. These references are:

US 12276847 B2

US 20230266546 A1

US 20210311272 A1

US 20210263235 A1

US 20210055490 A1

US 20200192040 A1

US 20190339458 A1

US 20180287702 A1

US 20170343738 A1

US 20150086211 A1

US 20150050019 A1

US 20140219615 A1

US 20130230283 A1

US 20130198414 A1

US 20120099822 A1

US 20100322554 A1

US 20100322562 A1

US 20100081303 A1

US 20070258683 A1

US 20070237470 A1

US 20060189220 A1

US 20040028344 A1

US 20030180012 A1

Contact Information

Any inquiry concerning this communication or earlier communications from the examiner should be directed to **Error! Unknown document property name.** whose telephone number is Error! Unknown document property name.. The examiner can normally be reached on 9-19.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Thomas Hollweg can be reached on571-270-1739. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see http://pair-direct.uspto.gov. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/KAVEH C KIANNI/
Primary Examiner, Art Unit 2874